ROCSCIENCE

Case12_MaxOutput
SLIDE - An Interactive Slope Stability Program
Rocscience Inc.
Date Created: 12/12/2016
Software Version: 9.041

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Slide2 Analysis Information

Case12_MaxOutput

Project Summary

File Name:	Case12_MaxOutput.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:08.626s
Project Title:	SLIDE - An Interactive Slope Stability Program
Author:	Rocscience Inc.
Company:	Rocscience Inc.
Date Created:	12/12/2016

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	50
Tolerance:	0.0001
Maximum number of iterations:	75
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

top layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	29.4 kPa
Phi	12 °
Water Surface	None
Ru Value	0

middle layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	9.8 kPa
Phi	5 °
Water Surface	None
Ru Value	0

lower layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	294 kPa
Phi	40 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	0.419809
Center:	26.967, 45.140
Radius:	31.450
Left Slip Surface Endpoint:	17.987, 15.000
Right Slip Surface Endpoint:	56.737, 35.000
Resisting Moment:	34638.4 kN-m
Driving Moment:	82509.8 kN-m
Total Slice Area:	350.585 m2
Surface Horizontal Width:	38.7497 m
Surface Average Height:	9.04742 m

Method: spencer

FS	0.422730
Center:	26.967, 45.140
Radius:	31.450
Left Slip Surface Endpoint:	17.987, 15.000
Right Slip Surface Endpoint:	56.737, 35.000
Resisting Moment:	34879.4 kN-m
Driving Moment:	82509.8 kN-m
Resisting Horizontal Force:	902.406 kN
Driving Horizontal Force:	2134.71 kN
Total Slice Area:	350.585 m2
Surface Horizontal Width:	38.7497 m
Surface Average Height:	9.04742 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	8943
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	4451
Number of Invalid Surfaces:	4492

Error Codes
Error Code -108 reported for 83 surfaces
Error Code -111 reported for 4409 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.419809

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.781771	5.34316	-15.8501	middle layer	9.8	5	26.326	11.0519	14.309	0	14.309	6.83464	6.83464
2	0.781771	16.117	-14.3747	middle layer	9.8	5	29.1999	12.2584	28.0995	0	28.0995	20.6159	20.6159
3	0.781771	26.5771	-12.9089	middle layer	9.8	5	31.9551	13.415	41.3199	0	41.3199	33.996	33.996
4	0.781771	36.7283	-11.4517	middle layer	9.8	5	34.5953	14.5234	53.9891	0	53.9891	46.9809	46.9809
5	0.781771	46.5757	-10.0019	middle layer	9.8	5	37.1244	15.5851	66.1245	0	66.1245	59.5772	59.5772
6	0.781771	56.1236	-8.55862	middle layer	9.8	5	39.5454	16.6015	77.7418	0	77.7418	71.7903	71.7903
7	0.781771	65.3757	-7.12078	middle layer	9.8	5	41.8613	17.5738	88.8546	0	88.8546	83.6251	83.6251
8	0.781771	74.335	-5.68743	middle layer	9.8	5	44.0746	18.5029	99.4749	0	99.4749	95.0854	95.0854
9	0.781771	83.004	-4.25764	middle layer	9.8	5	46.1874	19.3899	109.613	0	109.613	106.174	106.174
10	0.781771	91.3846	-2.83051	middle layer	9.8	5	48.2014	20.2354	119.277	0	119.277	116.894	116.894
11	0.781771	99.4781	-1.40514	middle layer	9.8	5	50.1185	21.0402	128.476	0	128.476	127.247	127.247
12	0.781771	107.285	0.0193669	middle layer	9.8	5	51.94	21.8049	137.217	0	137.217	137.234	137.234
13	0.781771	114.807	1.44388	middle layer	9.8	5	53.6665	22.5297	145.501	0	145.501	146.854	146.854
14	0.781771	122.041	2.86929	middle layer	9.8	5	55.2997	23.2153	153.337	0	153.337	156.109	156.109
15	0.781771	128.989	4.29648	middle layer	9.8	5	56.8394	23.8617	160.726	0	160.726	164.996	164.996
16	0.781771	135.649	5.72635	middle layer	9.8	5	58.2865	24.4692	167.67	0	167.67	173.515	173.515
17	0.781771	142.017	7.15981	middle layer	9.8	5	59.6409	25.0378	174.17	0	174.17	181.662	181.662
18	0.781771	148.094	8.59779	middle layer	9.8	5	60.9032	25.5677	180.225	0	180.225	189.433	189.433
19	0.781771	153.874	10.0413	middle layer	9.8	5	62.0723	26.0585	185.836	0	185.836	196.827	196.827
20	0.781771	159.354	11.4912	middle layer	9.8	5	63.1485	26.5103	191.001	0	191.001	203.838	203.838
21	0.781771	164.531	12.9486	middle layer	9.8	5	64.1311	26.9228	195.714	0	195.714	210.46	210.46
22	0.781771	169.399	14.4147	middle layer	9.8	5	65.0189	27.2955	199.974	0	199.974	216.685	216.685
23	0.781771	173.951	15.8904	middle layer	9.8	5	65.8109	27.628	203.774	0	203.774	222.509	222.509
24	0.781771	178.183	17.377	middle layer	9.8	5	66.5057	27.9197	207.109	0	207.109	227.921	227.921
25	0.781771	182.085	18.8759	middle layer	9.8	5	67.1019	28.17	209.97	0	209.97	232.913	232.913
26	0.781771	185.649	20.3882	middle layer	9.8	5	67.5976	28.3781	212.349	0	212.349	237.472	237.472
27	0.781771	188.866	21.9156	middle layer	9.8	5	67.9904	28.543	214.234	0	214.234	241.588	241.588
28	0.781771	191.724	23.4595	middle layer	9.8	5	68.278	28.6637	215.613	0	215.613	245.243	245.243
29	0.781771	194.212	25.0218	middle layer	9.8	5	68.4571	28.7389	216.473	0	216.473	248.426	248.426
30	0.781771	196.315	26.6041	middle layer	9.8	5	68.5245	28.7672	216.795	0	216.795	251.116	251.116

31	0.781771	198.018	28.2087	middle layer	9.8	5	68.4761	28.7469	216.565	0	216.565	253.295	253.295
32	0.781771	199.303	29.8379	middle layer	9.8	5	68.308	28.6763	215.758	0	215.758	254.938	254.938
33	0.781771	200.149	31.494	middle layer	9.8	5	68.0147	28.5532	214.35	0	214.35	256.019	256.019
34	0.781771	200.533	33.1801	middle layer	9.8	5	67.5905	28.3751	212.315	0	212.315	256.511	256.511
35	0.781771	200.429	34.8993	middle layer	9.8	5	67.0288	28.1393	209.619	0	209.619	256.378	256.378
36	0.781771	199.805	36.6553	middle layer	9.8	5	66.3216	27.8424	206.226	0	206.226	255.58	255.58
37	0.781771	198.627	38.4524	middle layer	9.8	5	65.4602	27.4808	202.092	0	202.092	254.073	254.073
38	0.781771	196.851	40.2955	middle layer	9.8	5	64.4338	27.0499	197.167	0	197.167	251.802	251.802
39	0.781771	193.006	42.1904	middle layer	9.8	5	62.9108	26.4105	189.859	0	189.859	246.884	246.884
40	0.781771	182.797	44.144	middle layer	9.8	5	59.9477	25.1666	175.641	0	175.641	233.824	233.824
41	0.781771	171.225	46.1647	middle layer	9.8	5	56.6848	23.7968	159.985	0	159.985	219.023	219.023
42	0.781771	158.789	48.2627	middle layer	9.8	5	53.2371	22.3494	143.441	0	143.441	203.115	203.115
43	0.781771	145.378	50.451	middle layer	9.8	5	49.5845	20.816	125.914	0	125.914	185.96	185.96
44	0.781771	130.852	52.7459	middle layer	9.8	5	45.7024	19.1863	107.286	0	107.286	167.379	167.379
45	0.781771	115.025	55.1693	middle layer	9.8	5	41.5594	17.447	87.4054	0	87.4054	147.133	147.133
46	0.781771	97.6441	57.7509	middle layer	9.8	5	37.1144	15.5809	66.0765	0	66.0765	124.901	124.901
47	0.697054	70.8967	60.37	top layer	29.4	12	64.2944	26.9914	-11.3317	0	-11.3317	101.709	101.709
48	0.697054	53.863	63.0559	top layer	29.4	12	54.6846	22.9571	-30.3115	0	-30.3115	77.2724	77.2724
49	0.697054	34.5891	66.0198	top layer	29.4	12	44.5017	18.6822	-50.4234	0	-50.4234	49.6221	49.6221
50	0.697054	12.1551	69.3861	top layer	29.4	12	33.6143	14.1116	-71.926	0	-71.926	17.4378	17.4378

Global Minimum Query (spencer) - Safety Factor: 0.42273

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.781771	5.34316	-15.8501	middle layer	9.8	5	27.2101	11.5025	19.4599	0	19.4599	11.7346	11.7346
2	0.781771	16.117	-14.3747	middle layer	9.8	5	30.2638	12.7934	34.2148	0	34.2148	26.4586	26.4586
3	0.781771	26.5771	-12.9089	middle layer	9.8	5	33.1612	14.0182	48.2146	0	48.2146	40.6142	40.6142
4	0.781771	36.7283	-11.4517	middle layer	9.8	5	35.9099	15.1802	61.4959	0	61.4959	54.2214	54.2214
5	0.781771	46.5757	-10.0019	middle layer	9.8	5	38.5169	16.2822	74.0924	0	74.0924	67.2995	67.2995
6	0.781771	56.1236	-8.55862	middle layer	9.8	5	40.9882	17.327	86.0335	0	86.0335	79.8648	79.8648
7	0.781771	65.3757	-7.12078	middle layer	9.8	5	43.3294	18.3166	97.3453	0	97.3453	91.9324	91.9324
8	0.781771	74.335	-5.68743	middle layer	9.8	5	45.545	19.2532	108.051	0	108.051	103.515	103.515
9	0.781771	83.004	-4.25764	middle layer	9.8	5	47.6392	20.1385	118.17	0	118.17	114.624	114.624
10	0.781771	91.3846	-2.83051	middle layer	9.8	5	49.6158	20.9741	127.721	0	127.721	125.268	125.268
11	0.781771	99.4781	-1.40514	middle layer	9.8	5	51.478	21.7613	136.719	0	136.719	135.456	135.456
12	0.781771	107.285	0.0193669	middle layer	9.8	5	53.2288	22.5014	145.177	0	145.177	145.195	145.195
13	0.781771	114.807	1.44388	middle layer	9.8	5	54.87	23.1952	153.109	0	153.109	154.492	154.492
14	0.781771	122.041	2.86929	middle layer	9.8	5	56.4043	23.8438	160.522	0	160.522	163.349	163.349
15	0.781771	128.989	4.29648	middle layer	9.8	5	57.8334	24.4479	167.426	0	167.426	171.771	171.771
16	0.781771	135.649	5.72635	middle layer	9.8	5	59.1583	25.008	173.829	0	173.829	179.761	179.761
17	0.781771	142.017	7.15981	middle layer	9.8	5	60.3809	25.5248	179.735	0	179.735	187.32	187.32
18	0.781771	148.094	8.59779	middle layer	9.8	5	61.5014	25.9985	185.15	0	185.15	194.449	194.449
19	0.781771	153.874	10.0413	middle layer	9.8	5	62.5212	26.4296	190.077	0	190.077	201.147	201.147
20	0.781771	159.354	11.4912	middle layer	9.8	5	63.44	26.818	194.517	0	194.517	207.413	207.413
21	0.781771	164.531	12.9486	middle layer	9.8	5	64.2585	27.164	198.472	0	198.472	213.246	213.246
22	0.781771	169.399	14.4147	middle layer	9.8	5	64.9765	27.4675	201.94	0	201.94	218.64	218.64
23	0.781771	173.951	15.8904	middle layer	9.8	5	65.5934	27.7283	204.921	0	204.921	223.594	223.594
24	0.781771	178.183	17.377	middle layer	9.8	5	66.1086	27.9461	207.411	0	207.411	228.099	228.099
25	0.781771	182.085	18.8759	middle layer	9.8	5	66.5217	28.1207	209.406	0	209.406	232.15	232.15
26	0.781771	185.649	20.3882	middle layer	9.8	5	66.8311	28.2515	210.902	0	210.902	235.74	235.74
27	0.781771	188.866	21.9156	middle layer	9.8	5	67.0352	28.3378	211.888	0	211.888	238.857	238.857
28	0.781771	191.724	23.4595	middle layer	9.8	5	67.1329	28.3791	212.36	0	212.36	241.494	241.494
29	0.781771	194.212	25.0218	middle layer	9.8	5	67.1216	28.3743	212.305	0	212.305	243.635	243.635
30	0.781771	196.315	26.6041	middle layer	9.8	5	66.9988	28.3224	211.712	0	211.712	245.268	245.268
31	0.781771	198.018	28.2087	middle layer	9.8	5	66.7615	28.2221	210.565	0	210.565	246.376	246.376
32	0.781771	199.303	29.8379	middle layer	9.8	5	66.4067	28.0721	208.85	0	208.85	246.94	246.94
33	0.781771	200.149	31.494	middle layer	9.8	5	65.93	27.8706	206.548	0	206.548	246.94	246.94

34	0.781771	200.533	33.1801	middle layer	9.8	5	65.327	27.6157	203.634	0	203.634	246.351	246.351
35	0.781771	200.429	34.8993	middle layer	9.8	5	64.5925	27.3052	200.085	0	200.085	245.144	245.144
36	0.781771	199.805	36.6553	middle layer	9.8	5	63.7203	26.9365	195.871	0	195.871	243.289	243.289
37	0.781771	198.627	38.4524	middle layer	9.8	5	62.7034	26.5066	190.957	0	190.957	240.749	240.749
38	0.781771	196.851	40.2955	middle layer	9.8	5	61.5331	26.0119	185.303	0	185.303	237.479	237.479
39	0.781771	193.006	42.1904	middle layer	9.8	5	59.9085	25.3251	177.453	0	177.453	231.757	231.757
40	0.781771	182.797	44.144	middle layer	9.8	5	56.9806	24.0874	163.305	0	163.305	218.608	218.608
41	0.781771	171.225	46.1647	middle layer	9.8	5	53.8029	22.7441	147.952	0	147.952	203.988	203.988
42	0.781771	158.789	48.2627	middle layer	9.8	5	50.4852	21.3416	131.922	0	131.922	188.511	188.511
43	0.781771	145.378	50.451	middle layer	9.8	5	47.0158	19.875	115.158	0	115.158	172.093	172.093
44	0.781771	130.852	52.7459	middle layer	9.8	5	43.3803	18.3381	97.5914	0	97.5914	154.631	154.631
45	0.781771	115.025	55.1693	middle layer	9.8	5	39.5612	16.7237	79.1388	0	79.1388	135.995	135.995
46	0.781771	97.6441	57.7509	middle layer	9.8	5	35.5376	15.0228	59.6968	0	59.6968	116.022	116.022
47	0.697054	70.8967	60.37	top layer	29.4	12	67.0856	28.3591	-4.89699	0	-4.89699	113.051	113.051
48	0.697054	53.863	63.0559	top layer	29.4	12	58.3647	24.6725	-22.2411	0	-22.2411	92.5828	92.5828
49	0.697054	34.5891	66.0198	top layer	29.4	12	49.3077	20.8438	-40.2536	0	-40.2536	70.5965	70.5965
50	0.697054	12.1551	69.3861	top layer	29.4	12	39.8902	16.8628	-58.9833	0	-58.9833	47.065	47.065

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.419809

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	17.9873	15	0	0	0
2	18.7691	14.778	23.7569	0	0
3	19.5508	14.5777	52.2144	0	0
4	20.3326	14.3985	84.5995	0	0
5	21.1144	14.2401	120.195	0	0
6	21.8961	14.1023	158.335	0	0
7	22.6779	13.9846	198.397	0	0
8	23.4597	13.887	239.801	0	0
9	24.2415	13.8091	282.002	0	0
10	25.0232	13.7509	324.489	0	0
11	25.805	13.7122	366.782	0	0
12	26.5868	13.6931	408.427	0	0
13	27.3685	13.6933	448.996	0	0
14	28.1503	13.713	488.084	0	0
15	28.9321	13.7522	525.307	0	0
16	29.7139	13.8109	560.303	0	0
17	30.4956	13.8893	592.725	0	0
18	31.2774	13.9875	622.246	0	0
19	32.0592	14.1057	648.556	0	0
20	32.8409	14.2442	671.357	0	0
21	33.6227	14.4031	690.37	0	0
22	34.4045	14.5829	705.326	0	0
23	35.1863	14.7838	715.974	0	0
24	35.968	15.0063	722.072	0	0
25	36.7498	15.251	723.395	0	0
26	37.5316	15.5183	719.73	0	0
27	38.3133	15.8088	710.877	0	0
28	39.0951	16.1234	696.65	0	0
29	39.8769	16.4626	676.877	0	0
30	40.6586	16.8275	651.403	0	0
31	41.4404	17.2191	620.086	0	0
32	42.2222	17.6384	582.806	0	0
33	43.004	18.0868	539.459	0	0
34	43.7857	18.5658	489.966	0	0
35	44.5675	19.077	434.274	0	0
36	45.3493	19.6223	372.358	0	0
37	46.131	20.2041	304.231	0	0
38	46.9128	20.8249	229.949	0	0
39	47.6946	21.4878	149.623	0	0
40	48.4764	22.1964	64.2654	0	0
41	49.2581	22.9552	-22.1376	0	0
42	50.0399	23.7694	-108.085	0	0
43	50.8217	24.6457	-192.162	0	0
44	51.6034	25.5924	-272.603	0	0
45	52.3852	26.6203	-347.156	0	0
46	53.167	27.7438	-412.87	0	0
47	53.9488	28.9829	-465.729	0	0
48	54.6458	30.2085	-407.025	0	0
49	55.3429	31.5798	-327.339	0	0
50	56.0399	33.1469	-217.302	0	0
51	56.737	35	0	0	0

Global Minimum Query (spencer) - Safety Factor: 0.42273

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	17.9873	15	0	0	0
2	18.7691	14.778	25.5924	3.83189	8.5155
3	19.5508	14.5777	56.1081	8.40094	8.51551
4	20.3326	14.3985	90.6727	13.5762	8.5155
5	21.1144	14.2401	128.486	19.238	8.51554
6	21.8961	14.1023	168.815	25.2762	8.51548
7	22.6779	13.9846	210.982	31.5899	8.51552
8	23.4597	13.887	254.364	38.0854	8.51552
9	24.2415	13.8091	298.384	44.6764	8.51552
10	25.0232	13.7509	342.507	51.2828	8.51551
11	25.805	13.7122	386.233	57.8299	8.51552
12	26.5868	13.6931	429.101	64.2484	8.51552
13	27.3685	13.6933	470.678	70.4736	8.51551
14	28.1503	13.713	510.559	76.4448	8.5155
15	28.9321	13.7522	548.366	82.1057	8.51552
16	29.7139	13.8109	583.747	87.4032	8.51552
17	30.4956	13.8893	616.371	92.2879	8.51551
18	31.2774	13.9875	645.926	96.7132	8.51552
19	32.0592	14.1057	672.124	100.636	8.51554
20	32.8409	14.2442	694.692	104.015	8.51554
21	33.6227	14.4031	713.376	106.812	8.5155
22	34.4045	14.5829	727.939	108.993	8.51553
23	35.1863	14.7838	738.16	110.523	8.51551
24	35.968	15.0063	743.836	111.373	8.51552
25	36.7498	15.251	744.778	111.514	8.51551
26	37.5316	15.5183	740.813	110.92	8.51549
27	38.3133	15.8088	731.783	109.568	8.51549
28	39.0951	16.1234	717.549	107.437	8.51551
29	39.8769	16.4626	697.988	104.508	8.5155
30	40.6586	16.8275	672.992	100.766	8.51554
31	41.4404	17.2191	642.476	96.1966	8.51552
32	42.2222	17.6384	606.374	90.791	8.51551
33	43.004	18.0868	564.64	84.5423	8.51551
34	43.7857	18.5658	517.257	77.4478	8.51552
35	44.5675	19.077	464.235	69.5089	8.51551
36	45.3493	19.6223	405.616	60.732	8.51551
37	46.131	20.2041	341.482	51.1294	8.51552
38	46.9128	20.8249	271.96	40.7201	8.51554
39	47.6946	21.4878	197.233	29.5313	8.51552
40	48.4764	22.1964	118.322	17.7161	8.51552
41	49.2581	22.9552	38.9613	5.83359	8.51551
42	50.0399	23.7694	-39.44	-5.90527	8.51552
43	50.8217	24.6457	-115.572	-17.3044	8.51554
44	51.6034	25.5924	-187.836	-28.1243	8.51552
45	52.3852	26.6203	-254.238	-38.0665	8.51551
46	53.167	27.7438	-312.224	-46.7486	8.51551
47	53.9488	28.9829	-358.409	-53.6638	8.51551
48	54.6458	30.2085	-305.643	-45.7633	8.51552
49	55.3429	31.5798	-234.458	-35.1048	8.51549
50	56.0399	33.1469	-137.006	-20.5136	8.51551
51	56.737	35	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	0, 3 96, 3 96, 35 72, 35 48, 35 24, 19 18, 15 0, 15
Material Boundary	24, 19 72, 35
Material Boundary	0, 3 96, 35